UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE O Level

MARK SCHEME for the November 2005 question paper

5090 BIOLOGY

5090/02 Paper 2 maximum raw mark 80

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Section A

1	**(a)**	bacterium/named bacterium	;	**[1]**
	(b)	specific pH read from graph	;	
		ref. to acid/low pH	;	
		sours/curdles the milk	;	
		flavours the cheese	;	**[max. 3]**
	(c)	stomach	;	
		stomach is acidic/ref. HC*l*	;	
		enzyme has optimum low pH AW	;	
		young mammals consume only milk	;	
		curdling increases surface area	;	**[max. 2]**
	(d)	**(i)** the use of vegetable oil/unsaturated fat (A v.v. for animal fat)	;	
		(ii) Any two from: obesity or described, named circulatory disorder, dairy allergies	;;	**[3]**
				[9]
2	**(a)**	A - alveolus/air sac	;	
		B - **capillary**	;	
		C – RBC/erythrocyte	;	**[3]**
	(b)	one mark each for statements 2 and 3, remove a mark for each incorrect	;;	**[2]**
	(c)	dissolves	;	
		diffuses	;	
		combines with haemoglobin/forms oxyhaemoglobin	;	
		ref. any structure/substance which O_2 passes through	;	**[max. 3]**
	(d)	more oxygen in air breathed out	;	
		Fe needed for haemoglobin	;	
		less haemoglobin formed	;	
		less oxygen absorbed by (red) blood (cells)	;	**[max. 3]**
				[11]

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3	**(a)**	**(i)**	chlorophyll ;		**[1]**
		(ii)	magnesium ;		**[1]**
	(b)	**(i)**	carbon dioxide/carbon(IV)oxide ;		
			photosynthesis ;		**[2]**
		(ii)	supplied with oxygen ;		
			for respiration ;		
			waste product/carbon dioxide removed AW ;		
			camouflage ;		
			food/part of food chain AW ;		
			importance in reproduction/eggs attached to leaves AW ;		
			shade ;		**[max. 4]**
	(c)		animal/fish + no CCW/vacuole/chloroplasts ;		**[1]**
					[9]
4	**(a)**	pollution ;			**[1]**
	(b)	**(i)**	carbon dioxide/carbon monoxide/sulphur dioxide/various oxides of nitrogen (R symbols) ;		**[1]**
		(ii)	global warming AW/carboxyhaemoglobin AW/acid rain (effect must be related to named gas) ;		**[1]**
	(c)	**(i)**	Any two from: drainage from land, ion/salt/nutrients or named, sewage, dung, warmer water ;;		**[2]**
		(ii)	bacteria in sewage/cow dung ;		
			decomposition AW ;		
			oxygen used up ;		
			bacteria + respiration ;		
			animals/plants + unable to respire ;		**[max. 4]**
					[9]
5	**(a)**	**(i)**	M - (inferior) **vena cava** ;		
			N - (systemic) **aorta** ;		
			O - pulmonary **vein** ;		**[3]**

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(ii) muscle ;

coronary + artery ; **[2]**

(b) **(i)** semilunar valves correctly shown + label ;

(ii) & **(iii)** tricuspid correctly shown ;

bicuspid correctly shown ;

both correctly labelled ; **[4]**

(c) 1, 4, 2, 3. (3 if all correct, 2 if correct sequence, but starting in wrong place, 1 for any two in sequence) ;;; **[3]**
atrium contracts then ventricle contracts, forcing tcv closed and as pressure builds up, slv open

[12]

The **maximum** mark for Section A is **50**

Section B

6 ref. in either **(a)** or **(b)** to reflex action ;

neurones/impulses ;

(a) **ciliary** ;

muscles + **relax** ;

(suspensory) ligaments ;

tighten AW ;

pull on lens AW ;

lens flatter AW ;

to focus on distant object/longer focal length ; **[max. 6*]**

(b) brighter light ;

circular iris muscles contract ;

radial muscles relax ;

pupil becomes smaller ; **[max. 4*]**

(* to include either of the first two marking points) **[Total = 10]**

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7 **(a)** oviduct/Fallopian tube ;

zygote ;

cell division/mitosis ;

ball of cells/blastula/blastocyst ;

uterus ;

spongy lining/endometrium ;

implantation ; **[max. 4]**

(b) **(i)** Each need must be qualified with the importance to pregnancy (max. 1 for list of three unqualified nutrients)

iron + blood production ;

calcium/phosphate + bone ;

protein for embryonic growth ;

vitamins for healthy development ; **[max. 3]**

(ii) contains correct proportions of dietary requirements ;

antibodies ;

correct temperature ;

inexpensive/readily available ;

sterile ; **[max. 3]**

[Total = 10]

8 E (A any point marked * to score once only – up to max. for each section.)

(a) water + inside cells* ;

osmosis* ;

pressure/turgor* ;

cells push against one another* ;

strengthening/lignin ;

in xylem ;

effect of roots ; **[max. 4]**

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(b) **(i)** stem droops/sags AW ;

(plus any points marked * from above)

(ii) insufficient water in plant ;

water lost by plant faster than it is absorbed ;

lack of water in soil ;

low humidity ;

wind ;

high temp ; **[max. 6]**

[Total = 10]

8 O

(a) **(i)** no roots to bind soil AW ;

no protection for soil from wind/rain ;

soil washed away ; **[3]**

(ii) less transpiration ;

fewer clouds ;

less rainfall ;

ref. to temperature (higher temperatures without trees) ; **[max. 3]**

(iii) loss of livelihood AW ;

loss of food ;

loss of fuel ;

loss of remedies ;

homelessness/relocation ; **[max. 3]**

[N.B. MAX. 8 for **(a)**]

(b) nutrients are in the seed ;

nutrients from the soil are not required ;

Any TWO from: needs water, oxygen, suitable temperature ;; **[max. 2]**

[Total = 10]

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